

Risks & Mitigation – Smart Irrigation System

This document lists potential risks in the system and suggested mitigation strategies.

1. ET Calculator Accuracy

Risk: Evapotranspiration (ET) calculations may be inaccurate due to sensor errors or incorrect parameters.

Mitigation:

- Use the **PyETo library** as reference for calculations.
 - Validate ET values against published agricultural data.
 - Perform periodic calibration of temperature, humidity, and solar sensors.
 - Log ET results and compare with historical trends for anomaly detection.
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2. Rule Engine Behavior

Risk: The rule engine may be too **conservative** (watering too little) or **aggressive** (watering too much).

Mitigation:

- Test rule engine using **historical sensor data**.
 - Adjust thresholds (soil moisture limits, ET levels) iteratively based on testing.
 - Implement fallback logic for missing or invalid sensor data.
 - Document and log each decision for auditing and analysis.
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3. Sensor Failures or Missing Data

Risk: Faulty or disconnected sensors can lead to incorrect irrigation decisions.

Mitigation:

- Implement **last-known-valid readings** in the rule engine.
 - Send alerts to dashboard when sensors are unresponsive.
 - Regularly check and maintain IoT sensors and wiring.
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4. Backend API or Database Downtime

Risk: If the backend API or database is unavailable, the system cannot process sensor data or make decisions.

Mitigation:

- Use **Docker restart policies** to auto-restart services.
- Monitor containers using logs and alerts.
- Ensure backup database and replication if needed.
- Schedule maintenance during low irrigation periods.

5. Weather Forecast Unavailability

Risk: Weather API failure may affect irrigation decisions (e.g., rain expected).

Mitigation:

- Rule engine can operate **without weather forecast** using soil moisture and ET only.
 - Retry weather API calls multiple times before fallback.
 - Log missing forecast events for review.
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6. Human Error

Risk: Manual intervention may override system settings incorrectly.

Mitigation:

- Include confirmation prompts for manual overrides.
- Keep detailed logs of all manual actions.
- Provide clear guidance in **Technician FAQ** for safe operations.